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Wildfire Progression Prediction and Validation Using Satellite Data and Remote Sensing in Sonoma, California

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***Abstract*—Wildfire has been a serious disaster in the United States for a long time with increasing devastation power, and occurrence risk. Compared to wildfire detection in which numerous kinds of research have been conducted, there is limited research on real-time wildfire progression. To bridge this gap, this paper proposes to develop a dynamic daily wildfire progression model using deep reinforcement learning, which can help to track and contain the wildfire. This paper focuses on wildfires that occurred in Sono1ma County for model development training and testing purposes. However, the proposed approach is not limited to Sono1ma County and can be applied to any location if the data is available. This research collected various input data, including weather data, land cover data, land fuel remote sensing data, and fire history data, for the proposed models. The proposed models contain three major components: the wildfire detection model, the land fuel vegetation data classification model, and the wildfire progression prediction model using deep reinforcement learning. The research conducts comparative experiments to evaluate various integrated deep learning models and different combinations of deep learning approaches. Specifically, the wildfire progression prediction model uses DQN with MLP, A2C with MLP, DQN with CNN, and an innovative model incorporated with the Recurrent neural network LSTM (long short-term memory) and deep reinforcement learning. The top-performing model yields an accuracy of 85.87% and an F1 score of 92.17%. Considering the importance and potential devastation caused by wildfires, this paper introduces an intelligent solution consisting of three interconnected models for wildfire detection, land fuel vegetation confirmation, and progression prediction. The approach relies primarily on Deep Learning Models to ensure high accuracy in detecting and predicting the spread of wildfire, making it applicable to real-world scenarios for effective wildfire management.**

***Index Terms*—wildfire, machine learning, remote sensing, reinforcement learning, satellite images**

I. Introduction

Wildfires have become a growing threat in the United States, resulting in severe devastation and increased risk. Factors such as climate change, prolonged drought, higher temperatures, and the accumulation of flammable materials due to fire suppression activities have contributed to the escalation of wildfires [1]. These wildfires not only claim lives but also cause extensive damage to properties and forests, resulting in substantial financial losses. Research suggests that approximately 90% of wildfires are caused by human activities [2]. With rising temperatures and dry conditions, the wildfire processes have become increasingly uncontrollable. In 2020 alone, there were 58,950 wildfires that consumed over 10.1 million acres of land, causing $16.5 billion in damages [3]. California experienced five of the largest fires in that year, putting over 2 million properties at risk [4]. Consequently, the analysis and prediction of wildfire progression have become crucial in mitigating the impact of these disasters.

Figure 1 illustrates the acres burned in California wildfires from 1987 to 2020, revealing that 2020 was the most devastating year with over 4 million acres burned. Given the persistent threat of wildfires in California, it is imperative to employ machine learning techniques and big data-based models to prevent large-scale fires and effectively combat them. The proposed model encompasses fire detection, confirmation, and progression prediction models, which aim to mitigate wildfire damage and support strategic decision-making in firefighting efforts.

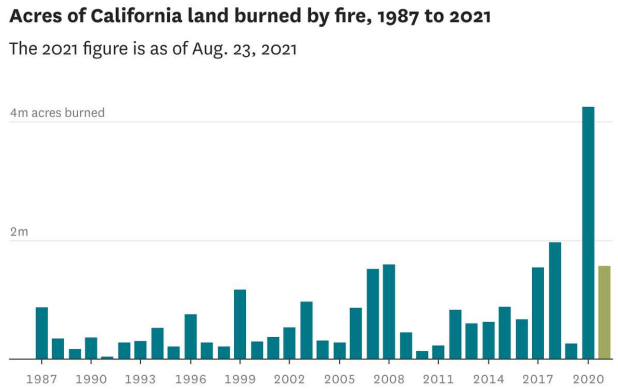

Fig. 1. Wildfire status in California[5].

Wildfire detection has been extensively researched, with var-

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ious algorithms implemented on data such as Normalized Difference Vegetation Index (NDVI), Land Surface Temperature (LST), and Thermal Anomalies (TA). However, these studies often overlook important parameters like soil condition, vegetation, and weather, and fail to address wildfire progression [6]. Existing research on wildfire progression detection often neglects critical factors like vegetation and current weather conditions [7]. One notable work by Radke et al. proposed the FireCast model, which used a 2D Convolutional Neural Network (CNN) to predict wildfire spread. Although this model simplified wildfire spread prediction, it did not consider important features such as vegetation data [8]. Another study by Huot et al. used a convolutional autoencoder to predict wildfire spreading the next day but lacked the ability to predict progression in shorter time frames [9].

To address these limitations and improve prediction accuracy, this paper aims to build a 3D deep reinforcement learning model for wildfire progression prediction. The proposed model utilizes complex data inputs such as weather data, land fuel vegetation indexes, land cover data, and wildfire history data. Reinforcement learning, deep reinforcement learning with feature extractions, and LSTM are employed to enhance prediction ability and accuracy. The models developed in this research achieve high accuracies [10].

This paper proposes a dynamic daily wildfire progression model using deep reinforcement learning. Initially, the model detects wildfires using satellite images and a Deep Learning Object Detection Model. If a fire is detected, the model progresses through various steps to track the wildfire's real-time progress, enabling containment in its early stages. The wildfire is treated as an agent, and its spread direction serves as the action, including east, south, west, north, southeast, northeast, southwest, northwest, or no movement. Rewards are calculated based on input data such as weather, fire history, land cover, and land fuel vegetation. The model aims to provide accurate daily predictions of how and where the wildfire will progress.

The structure of the paper is organized as follows: Section 2 provides an extensive literature survey and discusses previous technologies used in the field. Section 3 describes the datasets used, including data sources, parameters, and data preprocessing. Section 4 presents the machine learning model approaches, including model architecture, training, and testing processes. Section 5 compares the results of all the machine learning models. Finally, Section 6 concludes the paper and discusses potential future work.

II. LITERATURE SURVEY

For the fire detection and progression analysis task, there are plenty works have done in three main areas: Fire Detection Mode, Vegetation Fuel Model, and Fire Progression Model.

A. *Fire Detection Model*

Fire detection is a crucial step in the task of predicting fire progression. A well-performed fire detection model elevates the ability to evaluate the fire spreading direction. Researchers have proposed several methods to conduct fire detection experiences and received promising results. [11] introduced a cost-effective precise fire detection CNN model for surveillance cameras, yet the false alarm rates are high. [12] proposed a novel ensemble method that is the combination of YoloV5 and EfficientNet. [13] used a pre-trained Inception-ResNet-v2 model on the ImageNet dataset, and incorporated OpenCV to achieve real-time detection purposes. Another experiment conducted in [14] includes data mining, big data, and remote sensing technique. [15] proposed a deep neural computing model with a structure of 3 hidden layers and is in use for creating forest fire susceptibility maps for Gia Lai province. In order to evaluate the model performance, [15] uses CA and Kappa scores, and the model gave a high global performance result. The majority of the papers used classification metrics, including accuracy, precision, specificity, and F1-score. Table I is the comparison table of various fire detection papers.

TABLE I
COMPARISON OF DETECTION PAPERS

Model	**CNN-GoogleNet**	**Yolov5**	**ResNetv2**	**ANN+SVM**	**Deep Neural Computing**
Ref	[34]	[35]	[36]	[37]	[38]
Year	2018	2021	2022	2019	2021
Dataset	CCTV	Collected Images	Collected Images	Satellite Based Image	GIS
Data Type	Video	Images	Images	NASA	Geodata
Big Data	Yes	Yes	Yes	Yes	Yes
Real-time	Yes	Yes	Yes	Yes	No
Evaluation Metrics	Classification metrics	False Positive Rate	Classification metrics	Classification metrics	CA and Kappa
Accuracy %	94.4	94.7	99	98	89.4
Precision %	82	70.5	100	97	-
Recall %	98	75.6	98	98	-
F1-score %	89	-	99	97	-
CA %	-	-	-	-	95.40
Kappa	-	-	-	-	0.908
Techniques	Deep Learning	Ensemble, Deep Learning	Transfer Learning	Remote Sensing, Deep Leaning	Deep Neural Computing

B. *Vegetation Fuel Model*

It is critical to assess the moisture and vegetation land fuel status in the wildfire progression model. Many wildfire detections and progression prediction research include vegetation land fuel data and use the processed data as inputs to the models, but research particularly in developing vegetation fuel models is also done.

The research paper by T. C. McCandless et al. proposed developing a gridded real-time Live and Dead Fuel Moisture Content (LFMC and DFMC) dataset to be used in the wildfire progression prediction system. This research used three datasets - MODIS, WRF-Hydro/National Water Model (NWM), and Derived predictors, as inputs to the machine learning models [16]. From NWM, this research only utilized accumulated total evapotranspiration, soil moisture content,

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and land use category. While this research implemented Multiple Linear Regression (MLR), Random Forest (RF), Gradient Boosted Regression (GBR), and Artificial Neural Network (ANN) to train and test the input data, Random Forest resulted in the lowest Mean Absolute Error on the test data for predicting both LFMC and DFMC. The RF method was implemented in a real-time system to output LFMC and DFMC maps and datasets daily, which can be utilized as inputs to the wildfire progression prediction system.

C. *Fire Progression Prediction Model*

Fire progression prediction is a major issue now. Table II outlines and compares some of the fire progression prediction research done previously.

Many previous studies incorporate diverse datasets for predicting wildfire spread. In the study by [17], the dataset includes wildfire images, state information (distance, angles), and belief maps obtained through filtering techniques. Radke et al. [8] utilize historical fire perimeters, Landsat8 satellite imagery, elevation data, and weather interpolation. F. Huot et al. [9] combine historical wildfire data with remote sensing sources such as topography, weather, drought, vegetation, and population density data. The study by Subramanian and Crowley [18] focuses on satellite images but does not consider additional features like vegetation moisture and terrain. In contrast to previous work, this paper incorporates all the datasets to train the model, particularly, we use Landsat8 data for analyzing purposes and use Modis satellite images for daily wildfire progression prediction.

Various models and techniques are employed in these surveys. [17] proposes the use of Kalman and particle filters for image filtering, deep reinforcement learning for training a neural network control system, stochastic modeling for wildfire prediction, and the deep Q-learning algorithm for the controller model. Radke et al. [8] introduce the FireCast model, which utilizes a 2D CNN to predict wildfire spread. F. Huot et al. [9] employ a convolutional autoencoder, a deep-learning neural network model. Subramanian and Crowley [18] compare different reinforcement learning models, highlighting the performance of A3C and MCTS in predicting wildfire spread using satellite images. However, in this paper, we utilize a combination of multiple deep reinforcement learning methods with different extractors, such as Leaky ReLU and CNN, to find the best performing model.

III. STUDY AREA AND DATASETS

The objective of this paper is to build a fire progression prediction model. To achieve this output, all the environmental components such as weather conditions, land fuel data, fire status, and smoke status are required as inputs for the progression prediction model.

A. *Data Source*

Different datasets are collected for each component of the study. For weather conditions, NOAA data is collected, providing hourly information on temperature, humidity, precipitation, and wind. Satellite images for fire status, smoke status, and land fuel indexes are obtained from various sources. MODIS satellite images offer higher resolution, although they are updated on a daily basis. On the other hand, GOES-18 satellite images provide more frequent updates but have a lower resolution that may not meet the detection requirements. The paper explores methods for improving the resolution of GOES-18 data, considering available online resources. However, further testing is necessary to determine the applicability of these methods to the specific needs of the study. As a starting point, the team will use MODIS images, but they are also exploring additional data options from satellite image sources.

To facilitate convenient access within the team, the downloaded data will be stored in Google Drive. For larger datasets that exceed the computer's capacity, the team will utilize Google Colab to share code that enables data access directly from the cloud. This approach ensures efficient collaboration and enables working with extensive datasets without overwhelming local computing resources.

Table III shows detailed datasets with targeted parameters and sources.

B. *Data Pre-processing*

*1) **Satellite Image Data**:* MODIS (Moderate Resolution Imaging Spectroradiometer) satellite images are commonly used for fire detection due to their ability to cover large areas at a relatively low resolution. This paper relies on daily updated MODIS images for the object detection model. Although the spatial resolution of MODIS images is lower compared to Landsat images, this paper compared both datasets and found that fire detection using MODIS images outperformed Landsat images. The MODIS images were obtained in various spectral bands, and two specific bands were used for the analysis. The Terra MODIS 7-2-1 corrected reflectance band helps distinguish burn scars from low vegetation, while the Aqua MODIS true color corrected reflectance band provides visually appealing images by reducing atmospheric effects.

Figure 2 in the paper presents a sample of the processed data that is stored in google drive for further processing.

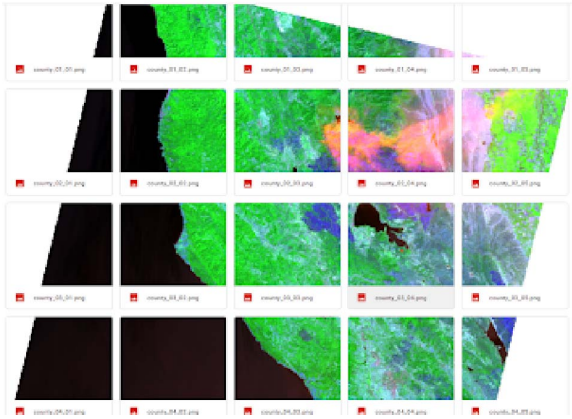

Fig. 2. Modis Data After Pre-processing

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TABLE II
COMPARISON OF PROGRESSION PAPERS

	Focused Problem	Dataset Used						Model Technique		Performance
		Fire History	Satellite	Remote Sensing	Weather	Vegetation	Landcover	Model Used	Time Interval	
[8]	Using deep learning to predict wildfire spreading	Y	Landsat8	Y	Y	Y	N	2D CNN	24 Hours	91.1% Recall
[9]	Next Day Wildfire spreading prediction based on remote sensing data	Y	N	Y	Y	Y	N	LR + RF	24 Hours	66.3% AUC(PR)
[17]	aircraft collected data to predict spreading	N	N	N	Y	N	N	DRL	N/A	35% ERROR
[18]	Using Spatial Reinforcement Learning to Build Forest Wildfire Dynamics Models From Satellite	N	Landsat 8	N	Y	N	N	A3C	16 Days	87.3%
	Proposed Model: using validation-based RL and DL to predict wildfire spreading daily	Y	Landsat 8 & Modis	Y	Y	Y	Y	QRDQN + MLP	24 Hours	92.17%

TABLE III
LIST OF DATASETS USED FOR OUR PROGRESSION MODELS

Data Name	Usage	Source
Landsat 8	Satellite imagery for land cover analysis	https://earthexplorer.usgs.gov
Elevation Products	Elevation data for terrain analysis	https://apps.nationalmap.gov/downloader/
GOES-18	Real-time monitoring	https://www.goes-r.gov/multimedia/dataAndImageryImagesGoes-18.html
MODIS	Active fire data for fire detection	https://www.earthdata.nasa.gov/learn/find-data/near-real-time/firms/mcd14dl-nrt https://www.earthdata.nasa.gov/learn/find-data/near-real-time/firms/active-fire-data
NOAA Weather Data	Hourly global weather data	https://www.ncei.noaa.gov/access/search/data-search/global-hourly
Fire History Data	Historical fire perimeters for analysis	https://frap.fire.ca.gov/frap-projects/fire-perimeters/ https://frap.fire.ca.gov/mapping/gis-data/
Landcover Data	Information on land cover types	https://land.copernicus.eu/global/products/lc

*2) **Remote Sensing Data**:* Google Earth Engine (GEE) is a widely used tool to access satellite images and remote sensing data. Most of the satellite images and remote sensing data can be easily accessed in GEE and Python/javascript can be used to manipulate the data before downloading. Since the majority of the imagery data requires huge storage space, being able to select and filter the data from GEE is very helpful for the paper. The team decided to collect both the "MODIS Vegetation Indices 16-Day L3 Global 250m"[20] and "LANDSAT/LC08/C01/T1_8DAY_NDVI" [21] as the source for the vegetation indexes to compare performance in the model. The MODIS dataset collects data with properties of low clouds, low view angle, and the highest NDVI/ EVI value. The MODIS land product dataset also has multiple layers - one vegetation layer, two quality layers, the MODS reflectance band 1 (red), band 2 (near-infrared), band 3 (blue), band 7 (midinfrared), and four observation layers [22]. This Landsat dataset collects data from Landsat 8 satellite which moves around the earth's surfaces every 8 days. Therefore, the data collected will have missing data sometimes due to the cloud or other limitations [23]. The valid range for the NDVI is between 0 to 1 and the Landsat NDVI data is already in the current range. The Modis NDVI data were normalized to the range before use.

*3) **Weather Data**:* Weather conditions such as wind speed, direction, and humidity level greatly impact the progression of wildfire. Hence, the weather data has been one of the key parameters for this paper and is collected from the National Centers for Environmental Information (NCEI) [24], which manages weather information for the National Oceanic and Atmospheric Administration (NOAA). The collected weather data includes features such as DewPointTemperature, DryBulbTemperature, Precipitation, RelativeHumidity, WetBulbTemperature, WindDirection, and WindSpeed. Each of these features contributes unique information about specific weather conditions. DewPointTemperature indicates the temperature at which the air becomes saturated with water vapor. DryBulbTemperature represents the temperature measured by a thermometer exposed to the air but shielded from radiation and moisture. Precipitation refers to any form of moisture that falls from the atmosphere, such as rain or snow. Relative humidity quantifies the amount of water vapor present in the air relative to the maximum amount it could hold at a given temperature. WetBulbTemperature is the lowest temperature that can be reached through evaporative cooling of a wet surface. By analyzing these weather parameters, the

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TABLE IV
PROCESSED DATASET STATISTIC

Dataset	Raw	Pre-processed	Transformed
Landsat	TIF	(6000,7)	(6000,7)
DEM	GeoTIFF	(4000,3)	(4000,3)
NDVI - MODIS	.csv	(80, 9)	(80, 9)
MODIS-Daily	.jpg	(250)	(234)
NDVI - Landsat	.csv	(139, 7)	(139, 7)
Weather	.csv	(3590, 40)	(3590, 9)

paper aims to gain insights into the atmospheric conditions during wildfire events

*4) **Fire History Data**:* Fire history data from 2006 to 2022 were obtained from the Fire and Resource Assessment Program (FRAP). Using QGIS, fires that occurred in Sonoma County were identified and extracted from the dataset. The extracted data was then divided into 1km by 1km grids, and the resulting grids were exported to a CSV file. The CSV file included information such as the grid's coordinates, fire name, containment date, start date, and a fire status column. The fire status column indicated whether a grid experienced a fire or not, with 0 representing no fire and 1 representing a fire. For the model presented in this paper, four recent fires (Kincade, Nuns, Pocket, and Tubbs) were selected for training.

Once all the collected data are cleaned and processed, the final shape of each data is shown in Table IV.

IV. MODEL DEVELOPMENT

This paper is influenced by the data pre-processing and modeling of the research "Wildfire Progression Analysis and Prediction Using Big Data and Reinforce Learning Approaches" by Phan, My, et al. [25], and this paper has proposed and added additional methods.

A. Fire and Smoke Detection Model

The first model presented in this paper focuses on Fire and Smoke Detection using Satellite Images. The team employs the YOLOv5 (You Only Look Once) deep learning model, implemented using TensorFlow, PyTorch, and Keras. YOLOv5 was chosen due to its speed, accuracy, and ability to perform well even with low-quality images.

YOLOv5 is a real-time object detection model and an evolution of previous versions of YOLO. It consists of three main components: Backbone, Neck, and Head. The Backbone utilizes a Convolutional Neural Network to extract features from the images at different levels. The Neck combines these features and aids in generalizing objects of various sizes. The Head applies bounding boxes, class labels, and prediction scores to generate the final results. YOLOv5 gets its name from the fact that it uses 1x1 convolutions for predictions, enabling efficient processing of images.

While YOLOv5 has multiple versions such as YOLOv5n, YOLOv5s, YOLOv5m, etc., the team opts to use YOLOv5s for detecting wildfire images. This version is well-suited for detecting small objects, which is important considering the size of the fires in the MODIS daily data used in the paper. These wildfire images are not of very high quality, making YOLOv5s the most suitable choice for accurate detection.

B. Fire/Smoke Confirmation Model (Using NDVI)

Once the Fire and Smoke detection model identifies the presence of a fire, the team will utilize vegetation analysis, specifically focusing on the Normalized Difference Vegetation Index (NDVI) and Enhanced Vegetation Index (EVI), for fire confirmation and severity assessment. To achieve this, the team will employ XGBoost Classification.

XGBoost (Extreme Gradient Boosting) is a powerful machine learning algorithm that operates similarly to gradient boosting trees. However, XGBoost is known for its speed and accuracy compared to traditional decision trees. It assigns weights to variables and iteratively improves the model by incorporating the misclassified predictions. This allows XGBoost to effectively classify data.

In the proposed model flow, the fire detection model will obtain NDVI values for nine sampling points within each grid cell. These NDVI values will then be passed to the trained XGBoost detection model, which will confirm the presence of a fire and provide a classification index value calculated based on the NDVI data. The classification index from the land fuel data will serve as input for the Deep Reinforcement Learning model for Fire Progression.

C. Fire Progression Model

To achieve the main goal of computing real-time wildfire progression prediction in this paper, the final model built is a Fire Progression Prediction Model using deep reinforcement learning. When a wildfire is detected, there are many factors that can accelerate or decelerate the spread of the wildfire. In this paper, factors such as weather conditions, landscape status, and vegetation land fuel status play significant roles to impact wildfire spread. Fig 3 shows the progression Model Workflow.

The occurrence and spread of wildfires require oxygen, fuels, and heat. Burning status indicates if the wildfire still exists, it can be burning or not. Weather conditions, such as temperature, humidity, precipitation, and wind speed and direction are considered in the wildfire progression model. While humidity and precipitation measure the moisture levels in the air, temperature quantifies the heat in the air. As the weather temperature increases and humidity and precipitation decrease, the chance of wildfire occurrence increases. The wind also provides the oxygen needed to accelerate the spread of wildfire. The wildfire is more likely to spread in the same direction as the wind direction and the speed of the wind affects the speed of the wildfire spread. Therefore, high temperature, low humidity, low precipitation, and high wind speed create an ideal weather condition to spread wildfire quickly.

Landscape status describes the topology of the wildfire environment, such as the elevation of the area, hills, lakes,

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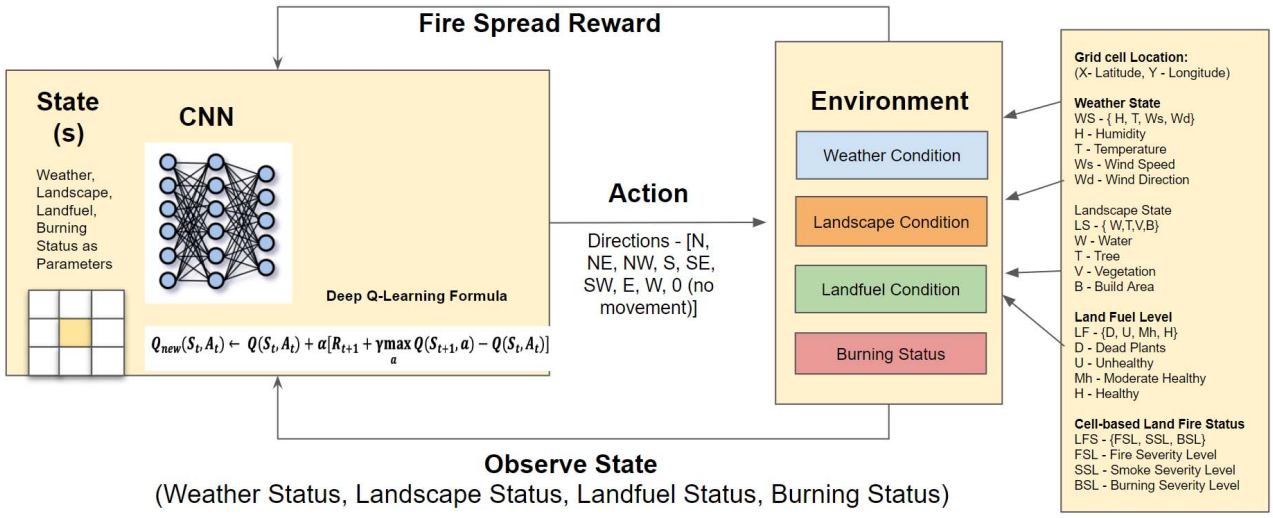

Fig. 3. Innovative Fire Progression Model Work Flow

and dry lands. For instance, if the wildfire is surrounded by a big lake, the lack of burning fuel and high moisture will stop the wildfire from spreading.

Land fuel conditions relates to the flammability of fuels present on the ground, such as grass, plants, fallen trees, dead plants, and leaves. Different types of vegetation have varying levels of moisture content, which affects their combustibility. Dead plants, dry leaves, and wood generally have lower moisture content and are more prone to ignition compared to living plants with higher moisture content.

Reinforcement learning is a trial-and-error learning process. Building a deep reinforcement learning model requires an agent, action, environment, state, and reward. In the fire progression prediction model, the agent is the fire, action is the direction the wildfire agent moves in the environment, which is the conditions, such as weather conditions, land fuel status, and land fire condition, and state is the status of each area or grid after the action. The reward is based on the reward policy after the action is taken.

Successful detection of wildfire is a trigger to start the fire progression prediction reinforcement learning model. The model algorithm can be separated into the below four steps.

Initially, when a wildfire is detected, the wildfire and its surrounding region's satellite image will form a grid. For example, the satellite image can be divided into a 3 by 3 grid. Depending on the satellite image resolution, the image may need to be divided further. The grid cell that the wildfire initially detected has a state So. The wildfire agent can move in different directions - North, North East, North West, South, South East, South West, East, or West, from the initial grid cell. Each grid cell that corresponds to the direction the wildfire can potentially move to have its own environment Ei, which comprises a different weather condition, land fire condition, land fuel status, burning status, and a state Si. The grid cell that the wildfire moves to has a state S1 and environment E1.

Secondly, to decide which direction or grid the fire moves to after a wildfire is detected, the weather condition, land fire condition, land fuel status, and fire status data of each of the surrounding grid cell's environment will be used as input to a reinforcement learning model. The reward policy is shown in Figure 4. For instance, under weather conditions, if the wildfire agent moves to a grid cell with no wind, which won't accelerate wildfire spread, the reward is 0, otherwise, the reward is 1. The agent will check the policy for each environment condition component for the best action. The agent tends to take action that results in the highest rewards. Within each grid cell, the reward of each environment component will then be added to become the total reward of the grid cell.

Additionally, as the total reward of each of the surrounding grid cells is calculated, each cell's reward will then be compared with the reward threshold that the paper set. The fire agent will move to the grid cell(s) whose total reward exceeds the reward threshold. The direction that the fire moves also depends on the wind direction. The maximum number of directions that the fire could move is three. For example, if the wind has a direction to the North, the three directions the wildfire can possibly move to are North, North East, and North West.

Lastly, after the wildfire agent moves to the new grid cell, it will check the current state and reward at the new cell. As the reward is greater than 0, indicating a potentially flammable condition, it will repeat the above three steps to decide the next direction and grid cell that the agent may move to. The grid cell that has a reward greater than 0 but less than the threshold indicates that it has a flammable condition and can potentially sustain fire. These cells will still be checked until no flammable condition remains, or the total reward equals 0.

This fire progression prediction model will use a few reinforcement learning models - Q-Learning, Deep Q-Learning

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Total Reward of Each Grid Cell:

$$G_r^i = \sum (W_r^i + L_r^i + F_r^i + B_r^i)$$

Reward	Scenario
Weather Condition (W_r^i)	
[-1,1]	If the agent moves to a grid that is opposite with the wind direction, reward 0. Otherwise, reward 1.
[0,1]	If the agent moves to a grid with no wind, reward 0. Otherwise, reward 1.
[-1,1]	If agent moves to a grid with low temperature reward -1. Otherwise, reward 1
[-1,1]	If agent moves to a grid with high humidity, reward -1. Otherwise, reward 1
[-1,1]	If agent moves to a grid high precipitation, reward -1. Otherwise, reward 1
Landscape Status (L_r^i)	
[-1,1]	If agent moves to a grid with water, reward -1. Otherwise, reward 1
[-1,1]	If agent moves to a grid with dry land (dessert) reward -1. Otherwise, reward 1
[0,1]	If agent moves to a grid that is uphill, reward 1. Otherwise, reward 0
[0,1]	If agent moves to a grid that is downhill, reward 0. Otherwise, reward 1
Land Fuel Status(F_r^i)	
[-1,1]	If agent moves to a grid with no vegetation, reward -1. Otherwise, reward 1
[-1,1]	If agent moves to a grid with high moisture content, reward -1. Otherwise, reward 1
Burning Status (B_r^i)	
[0,1]	If agent moves to a grid with fire, reward 0.5. Otherwise, reward 1.
[0,1]	If agent moves to a grid that is burnt, reward 0. Otherwise, reward 1.

Fig. 4. Innovative Reward Policy

model, and Proximal Policy Optimization.

Q-Learning is an off-policy temporal control algorithm. The current state has Q value Q(St, At), Rt+1 is the current reward, max Q(St+1, a) is the maximum Q value after taking action a, and this will also multiply by a discount factor. Discount factor * max Q(St+1, a) - Q(St, At) calculates the temporal difference error and by adding the reward Rt+1, the term calculates the temporal difference target estimate and also discounts or multiplies by the learning rate alpha. The Q-Learning enables the fire agent to take actions that result in the highest reward.

Proximal Policy Optimization is a policy gradient method proposed by the Openai team in 2017. PPO algorithm "uses fixed-length trajectory segments. In each iteration, each of N (parallel) actors collects T timesteps of data. Then we construct the surrogate loss on these NT timesteps of data and optimize it with minibatch SGD for K epochs"[28]. It utilizes multiple stochastic gradient ascents epochs on every policy update. PPO inherent the advantages of trust region policy optimization, but have better empirical complexity, easier implementation, and are more general.

Deep Q-Learning is similar to Q-learning with the Deep Neural Network approach. Each of the components - weather condition, landscape status, fuel status, and burning status parameters will not only go through a deep neural network, but the component also passes reinforcement learning rules. Instead of iteratively calculating the Q-values to update the Q-tables, Deep Q-learning uses deep neural to estimate the Q-values for state-action pairs at each given environment, and the Deep Q network (DQN) will approximate the optimal Q-function.

Convolutional Neural Network (CNN) is used as one of the feature extraction techniques that combine with the Deep Reinforcement Learning models to predict fire progression. CNN is one of the most commonly used deep learning models, and it was used to extract features to represent the state of the environment in a more meaningful way by learning the associated weights for each feature, such as the land fuel data, fire status data, weather condition data, and land cover data, to calculate and assign feature importances. After CNN feature extraction, these features are passed to the deep reinforcement learning model to represent the state of the environment.

A2C (Advantage-Actor Critic) is a type of deep reinforcement learning algorithm that combines the actor-critic algorithm with the advantage function. A2C works mainly by two factors: Actor and Critic. It combines the ideas of policy gradient methods and value-based methods. The actor takes actions in the environment (policy-based method), while the critic helps in evaluating the actions of the actor and provides feedback (value-based). The advantage function measures how well the actor at a particular stage is by comparing it to the average action[29].

To better memorize the features and rewards in each state in a time series manner with the reinforcement learning model, the team will implement Recurrent Network work in the reinforcement learning model state. Specifically, the RNN model to be used in this paper is the LSTM model which is introduced to solve the long-term dependencies on the model by allowing the passing of relevant inputs and forgetting the irrelevant information. The output memory cells in the LSTM network are inputs to the next memory cells. There is an input gate, a forget gate, and an output gate. The input data will go through the input gate, forget gate with multiple sigmoid and tanh functions, and outputs the relevant information in the output gate[30].

V. Result

THis paper is composed of two major parts. The fire detection model is to detect if there is a fire or not in each grid set. The progression model is to predict which is the next grid the fire will move to. To evaluate these models, classification reports and AUC/ROC curves are used to compare different models' performances.

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A. *Fire Detection Model*

The fire detection model was built using YOLOv5s. The model was trained using a cloned YOLOv5 repository. The model was trained using a .yml file exported from Roboflow. The training performance with the number of epochs used. The team used 100 epochs for this data. Precision and Recall are highest close to 50 epochs. The training loss is also close to zero when approaching 100 epochs. Fig.5 shows the testing image with fire detection confidence.

Fig. 5. Test image 1 with fire detection and confidence.

B. *Remote Sensing Detection (NDVI) Results*

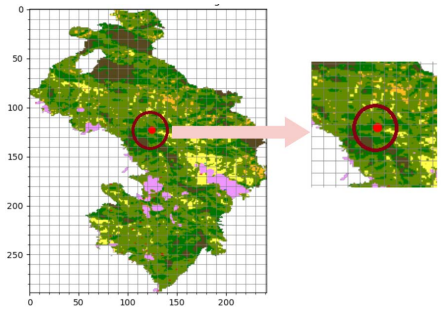
This model is to confirm fire by sampling the NDVI value around the area where the fire was detected. The training and testing dataset was upsampled to ensure adequate data for modeling. The model is trained by the training dataset and tested on the testing where the results return 100% accuracy. When the model is validated with a new dataset, the accuracy is 84%. Since the validating dataset is relatively small. For the next step, we would expand the size of the training, testing, and validating dataset to improve the performance and accuracy of the model.

C. *Fire Progression Model*

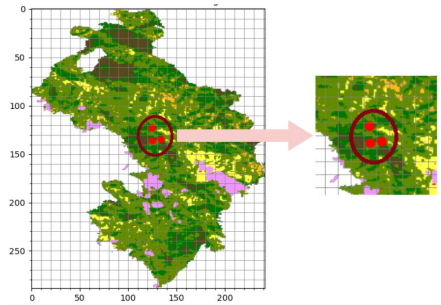
*1) **SARSA**:* There are four environments which are land cover, weather, land fuel status, and burning status that are set up for the SARSA model. The learning of the environment updates the 4 q-tables based on the reward table. After all the updating of the q-tables, the model is tested with Kincade Fire for validation and testing. It returned three new location detections with fire. When validating with the validation model, it showed the three locations were false positives. Therefore the SARSA model results are not reliable for fire progression analysis.

*2) **Q learning** :* For the Q learning model, the environment components were first set up based on the action reward table. Based on the location, date, and time that were passed in the functions, the closest location will be checked for the next step then get the score for the new location. The iteration runs for 10 episodes and gets different scores based on the location's latitude and longitude.

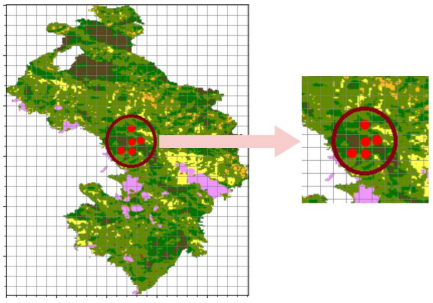
After the environments are set up, the agents in each of the environments will need to be trained and each of the q-tables



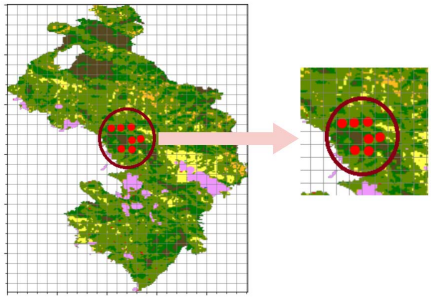
(a) Initial Fire Detected



(b) Progression predicted on the second day



(c) Progression predicted on the third day



(d) Progression predicted on the fourth day

Fig. 6. Progression Prediction for Kincade Fire With Landcover Background

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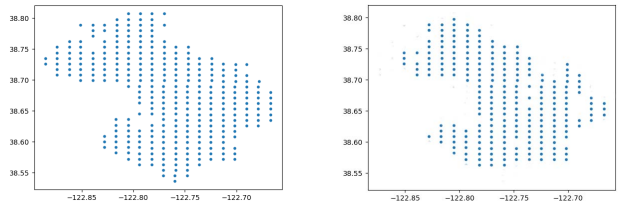

(a) Actual Kincade Fire Shape (b) Predicted Kincade Fire Shape

Fig. 7. Progression Prediction and Actual Result Comparision for Kincade Fire

will be updated. Most environments use daily datasets while the weather dataset is updated hourly. The q-table updates take a long time since each agent needs to be trained and the q-tables are updated every time the agents get trained. Since multiple environments and agents were being trained with big datasets, there are limitations when running the program. Colab was disconnected after running for a continuous 12 hours and required a restart and reconnecting to the program. The q-table updates could not be completed in the timeframe before Colab got disconnected. When running the validating and testing set against the incomplete q-table, the results returned minimal accuracy. Therefore, the Q-Learning model was not used in the final paper.

*3) **Deep Q-learning**:* The team constructed four baseline models (DQN, A2C, PPO, and QRDQN) with three different feature architects, MLP, CNN with ReLu activation, and CNN with Leaky ReLu activation. There is also an innovative model to incorporate a Recurrent neural network LSTM (long short-term memory) into the feature extraction and MLP.

After the initial fire is detected, the model utilizes parameters such as weather, land cover, historical fire data, vegetation, and wind direction to make a prediction. The resulting prediction is shown in Figure 6, where the red dots in the circles indicate the location of the fire. Figure 7 presents the prediction result compared with the actual burned area.

Evaluation and validation is done based on NDVI value. The team evaluated the model by comparing the observation space of the predicted new grid with the true grid cell. If the new grid cell landed on the on-fire true grid cell, it is a true positive. If the new grid cell landed on a no-fire grid cell, that means the model returns a false positive. If the new grid cell stops as determined there is no fire when the grid cell has fire, this means the model returns a false negative. As discussed above in section 4.4, accuracy was calculated using the sum of the true positives and true negatives divided by the total population, and F1 score measures the ratio of true positive detection to the overall positive detection. In general, the higher accuracy and F1 score, the better the model performs. The result is recorded in Table V.

For the four base models and three feature extraction combinations, DQN with MLP, A2C with MLP, and DQN with CNN model with ReLu active function returns the highest accuracy and F1 score 85.87% and 92.17%. PPO with MLP, QRDQN with MLP, and QRDQN with CNN with ReLu activation function followed closely by having accuracy scores of 85.86%, 85.86%, and F1 scores of 92.17%, and 92.15%.

TABLE V
MODEL RESULT COMPARISON

	Accu-racy	F1 score	Buffer time	run time
DQN + MLP	85.87%	92.17%	20000	3 h
A2C + MLP	85.87%	92.17%	20000	3 h
PPO + MLP	85.86%	92.17%	20000	3 h
QRDQN + MLP	85.85%	92.15%	20000	3 h
DQN + CNN (ReLu)	85.87%	92.17%	10000	3 h
A2C + CNN (ReLu)	85.64%	92.02%	10000	3 h
PPO + CNN (ReLu)	85.64%	92.02%	10000	3 h
QRDQN + CNN (ReLu)	85.85%	92.15%	10000	3 h
DQN + CNN (Leaky ReLu)	85.63%	92.02%	10000	3 h
A2C + CNN (Leaky ReLu)	85.64%	92.03%	10000	3 h
PPO + CNN (Leaky ReLu)	85.63%	92.03%	10000	3 h
QRDQN + CNN (Leaky ReLu)	85.63%	92.03%	10000	3 h
Recurrent PPO + MLPLSTM	85.63%	91.98%	20000	3 h

The recurrent PPO model uses a Recurrent neural network LSTM (long short-term memory) as the feature extraction. The feature extraction used in the model constructs LSTM together with MLP to add in the time series components in remembering the feature. The model accuracy returns as 85.63% and the F1 score as 91.98% respectively.

VI. CONCLUSION

In conclusion, this paper incorporates significant features that can influence wildfire progression and utilizes complex datasets including satellite images, remote sensing data, wildfire history, weather, and land cover information. The proposed wildfire progression prediction system comprises three main components: the wildfire detection model, wildfire risk assessment, and wildfire progression prediction model. The wildfire detection model utilizes object detection (YOLOv5) to accurately locate wildfires. The wildfire risk assessment incorporates land fuel vegetation indexes to evaluate the risk of wildfire occurrence and spread. The wildfire progression prediction models employ various deep reinforcement learning approaches, such as DQN with MLP, A2C with MLP, DQN with CNN, and a novel model incorporating LSTM and deep reinforcement learning. The highest-performing model achieves an accuracy of 85.87% and an F1 score of 92.17%. Although the training focuses on wildfires in Sonoma County,

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the high accuracy of the wildfire progression model indicates its potential for real-world applications beyond this specific location.

For future studies, we will incorporate elevation data to consider terrain factors in wildfire progression prediction. Additionally, as the resolution of GOES satellite images continues to improve, our research will focus on enhancing the models to provide real-time wildfire progression predictions, preferably on an hourly basis. We have collected GOES-18 images and are currently working on the training phase to leverage this dataset. By incorporating elevation data and improving the temporal resolution of satellite imagery, our goal is to enhance the accuracy and timeliness of wildfire progression prediction models, enabling their practical use in real-world scenarios.

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